

Calder N. Robbins

📞 (413) 992-8113 | ✉️ robbins.cal@northeastern.edu | 🔗 linkedin.com/in/calder-robbins | 🌐 github.com/cnrobbins

Summary | PhD candidate with a background in biomechanics, neuromechanical simulation, and assistive robotics. I develop exoskeleton hardware and controllers, as well as simulation-based frameworks for the development of device design and adaptive control strategies to enhance the performance of assistive devices.

Education

Northeastern University <i>PhD in Mechanical Engineering</i>	
University of Massachusetts Amherst <i>MS in Mechanical Engineering</i>	GPA: 3.9
<i>BS in Biomedical Engineering Cum laude</i>	GPA: 3.8

Skills

Programming Python, MATLAB, Simulink	Manufacturing 3D Printing (FDM, SLA, SLS)
Modeling SolidWorks, MuJoCo	Design Adobe Suite
Experimental Motion Capture, EMG, OpenSim, MyoSuite	

Publications

- Price, M., **Robbins, C.**, Szemethy, S., Abdikadirova, B., Olson, G., Hoogkamer, W., & Huber, M. E. (2025). *Adjustable Compliance Footwear Technology to Investigate Gait Adaptation*, IEEE Robotics and Automation Letters, doi: 10.1109/LRA.2025.3629969.
- **Robbins, C.***, Son, H.*, & Song, S. *MyoAssist 1.0: A MyoSuite module for simulating and training physical human–device interaction. **Manuscript in preparation***
- **Robbins, C.**, Grimmitt, A., Price, M., & Hoogkamer, H. *Effects of steep slopes and footwear bending stiffness on joint kinematics during walking. **Manuscript in preparation.***

Research Experience

NeuMove Lab <i>Graduate Research Assistant, PhD Candidate</i>	Aug 2024 – Present <i>Advisor: Seungmoon Song, Ph.D.</i>
– Project Lead on <i>MyoAssist</i>: An open-source platform for neuromechanical simulation and assistive robotics development. – Modeled and integrated exoskeleton and prosthetic devices within the MyoSuite simulation suite, and developed a control optimization framework for devices. – Presented a four-hour, hands-on tutorial workshop introducing <i>MyoAssist 1.0</i> at the 2025 <i>American Society of Biomechanics</i> meeting. – Presented <i>MyoAssist 0.1: Myosuite for Dexterity and Agility in Bionic Humans</i> at the 2025 <i>International Conference On Rehabilitation Robotics (ICORR)</i>. – Presented independent research on simulation-based exoskeleton control optimization at the 2025 <i>American Society of Biomechanics</i> meeting. – Actively developing novel exoskeleton hardware in collaboration with Professor Max Shepherd.	
Human Robot Systems Lab <i>Research Assistant, Graduate Research Assistant</i>	Feb 2022 – Aug 2024 <i>Advisor: Meghan Huber, Ph.D.</i>

Variable Stiffness Shoes

- Co-developed pneumatic variable-stiffness footwear capable of independently modulating midsole compliance under each foot for gait perturbation experiments.
- Designed and manufactured a novel adjustable "air pocket" for sneaker outsoles.
- Prototyped advanced footwear technology (AFT) through various benchtop methods and manufacturing processes (FDM, SLS, SLA 3D printing).
- Led hardware characterization of stiffness, damping, and step-response behavior.
- Conducted pilot studies demonstrating asymmetric compliance perturbations during walking to study gait adaptation and motor learning.

General

- Assisted with a study on exoskeleton assistance entrainment.
- Wrote "air pocket" manufacturing guides and best practices.

Uphill Biomechanics

- Designed and built a specialized uphill treadmill.
- Designed and ran a study on the effects of bending stiffness on the biomechanics of inclined walking.
- Wrote custom MATLAB and Python data processing and analysis pipelines.
- Presented work on steep-incline biomechanics at the 2024 *American Society of Biomechanics* meeting.

General

- Completed MATLAB data analysis for Asymmetrical footwear and split-belt running study of gait and balance for stroke and neuromuscular disease patient rehabilitation.
- Assisted with metabolic and biomechanics data collection for various studies.

Relevant Course Work

Human Movement Simulation**Fall 2025**

- Simulated variable-stiffness ankle prosthesis in MyoAssist with a co-evolutionary optimization framework to investigate how different gait cost priorities, such as metabolic effort, symmetry, and joint loading, shape personalized stiffness adaptation.

Wearable Robotics**Fall 2024**

- Implemented and compared CMA-ES and deep reinforcement learning frameworks in neuromechanical simulation to optimize bilateral ankle exoskeleton assistance.

Musculoskeletal Modeling and Simulation**Spring 2024**

- Simulated ankle exoskeleton in OpenSim Moco to assess the impact of post-stroke plantar flexion weakness on optimal assistance.

Human Robot Systems**Fall 2022**

- Simulated hip exoskeleton using OpenSim Moco to assess the impact of ground angle on optimal hip torque profiles during walking.
- Completed projects on optimal control, multi-DOF control and coordination, and impedance control.

Organizations

NEU MIE PhD Council | Member, Vice President**2024-Present**

- Organize and plan community building and outreach events for MIE PhD students, including seminars, networking events, and student-run activities.

UMass BME Diversity, Equity, & Inclusion Committee | Member**2021-2023**

- Help organize and plan community building and outreach events for the BME department.
- Awarded the **Biomedical Engineering Departmental Impact Award**.

UMass Fencing | President, Captain**2019-2021, 2023-2024**

- Organized and managed team activities, practices, and finances.
- Wrote yearly budget requests and allocated funds accordingly.

UMass Club Sports Executive Board | President**2020-2021**

- Advised all of UMass club sports on finances, scheduling, and general conflicts.
- Attended both Student Government and Organizational meetings with university leadership.

Work Experience

Beaver-Visitec International (BVI Medical)**Summer 2023***Research and Development Intern*

- Completed Medical Device Regulation (MDR) technical documentation and DHF remediation for multiple device families.
- Completed design reviews and all supporting documentation (User Needs, Product Requirements, Product Requirement Derivation, Design Verification Master Plan and Protocol, and Test Methods) for multiple devices.
- Completed both User Failure Mode and Effects Analysis (UFMEA) and Design Failure Mode and Effects Analysis (DFMEA) for multiple devices.
- Utilized 3D printing (FDM and SLS) to create fixtures and prototypes for device testing.

Pleasant Bay Community Boating**2019-2022***Sailing and Adaptive Sailing Coordinator*

- Accessible sailboat research, design, and construction for individuals with disabilities or limited mobility.
- Wrote sailing curriculum for a wide range of ages and skill levels.
- Managed a team of 20 sailing instructors.